

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

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Abstract—The rapid growth of supply chain networks and logistics operations has increased the need for efficient monitoring and performance analysis systems. Traditional reporting methods often fail to provide timely and actionable insights, leading to delays, inefficiencies, and suboptimal decision-making. This paper presents the design and development of an interactive Logistics Performance Dashboard that enables comprehensive analysis of shipping operations across multiple dimensions, including region, state, route, and shipping mode.

The proposed system utilizes data preprocessing, feature engineering, and exploratory data analysis techniques to transform raw logistics data into meaningful insights. Key performance indicators such as shipping lead time, route efficiency, and regional performance are analyzed and visualized using advanced graphical representations, including heatmaps, histograms, line charts, and choropleth maps. The dashboard is implemented using modern data visualization tools, providing an intuitive and interactive interface for users to explore data dynamically through filters such as date range and geographic region.

The system identifies critical patterns and trends in logistics performance, such as variations in delivery times across different states and shipping modes, as well as the presence of regional bottlenecks. By highlighting both high-performing and underperforming areas, the dashboard supports data-driven decision-making and helps organizations optimize their logistics strategies. Additionally, the comparative analysis of shipping modes enables a better understanding of trade-offs between speed and operational efficiency.

The results demonstrate that the proposed dashboard significantly improves visibility into logistics operations and enhances the ability to detect inefficiencies and delays. Compared to conventional static reports, the interactive nature of the system allows for faster analysis, improved user engagement, and more informed decision-making. Although the current system is limited to historical data analysis, it provides a strong foundation for future enhancements such as real-time monitoring, predictive analytics, and optimization techniques.

In conclusion, the Logistics Performance Dashboard serves as an effective tool for analyzing and improving supply chain performance. By integrating data analytics and visualization, the system contributes to more efficient logistics management and supports the transition toward intelligent, data-driven supply chain operations.

Index Terms—Logistics Dashboard, Supply Chain Analytics, Data Visualization, Streamlit, Route Efficiency, Shipment Analysis.

I. INTRODUCTION

In recent years, the rapid expansion of global trade and e-commerce has significantly increased the complexity of logistics and supply chain operations. Efficient management of transportation, delivery routes, and distribution networks has become critical for organizations aiming to meet customer expectations and maintain competitive advantage. However, traditional logistics management approaches often rely on static reports and manual analysis, which limit the ability to quickly identify inefficiencies, delays, and operational bottlenecks.

With the growing availability of data, there is a strong need for intelligent systems that can analyze logistics performance in a structured and meaningful way. Data-driven decision-making has emerged as a key enabler in optimizing supply chain processes, allowing organizations to gain deeper insights into their operations. In this context, data visualization and analytics tools play a crucial role in transforming large volumes of raw data into actionable information.

This paper presents the development of a Logistics Performance Dashboard designed to provide a comprehensive and interactive platform for analyzing shipping operations. The system focuses on evaluating key performance indicators such as shipping lead time, route efficiency, regional performance, and ship mode comparison. By leveraging data preprocessing techniques and visualization methods, the dashboard enables users to explore patterns and trends across different geographical regions and time periods.

One of the primary challenges in logistics analysis is the lack of visibility into performance variations across different operational dimensions. For instance, certain routes may consistently perform well, while others experience frequent delays due to factors such as distance, infrastructure, or demand fluctuations. Similarly, shipping performance may vary significantly across regions, states, and transportation modes. Addressing these challenges requires a system capable of integrating multiple data perspectives and presenting them in an intuitive manner.

The proposed dashboard addresses these challenges by providing interactive visualizations such as heatmaps, histograms, line charts, and geographic maps. These visual tools help in

identifying regional bottlenecks, analyzing temporal trends in shipping performance, and comparing the efficiency of different shipping modes. Additionally, the inclusion of filtering mechanisms allows users to customize their analysis based on specific regions and time frames, thereby enhancing usability and flexibility.

The objective of this work is to develop a system that not only improves visibility into logistics performance but also supports data-driven decision-making. By enabling users to identify inefficiencies and optimize operational strategies, the dashboard contributes to improved supply chain efficiency and overall business performance.

In summary, the proposed Logistics Performance Dashboard demonstrates the potential of integrating data analytics and visualization techniques in modern logistics management. It provides a scalable and user-friendly solution for analyzing complex logistics data and lays the foundation for future enhancements such as predictive analytics and real-time monitoring.

II. LITERATURE REVIEW

In recent years, the application of data analytics and visualization techniques in supply chain management has gained significant attention. Traditional logistics systems primarily depend on static reporting methods, which often lack real-time interactivity and limit the ability to identify operational inefficiencies. Studies have highlighted that such approaches make it difficult to monitor key performance indicators (KPIs) such as delivery time, route efficiency, and regional performance.

To address these limitations, several researchers have proposed the use of interactive dashboards for logistics analysis. These systems utilize data visualization techniques, including heatmaps, bar charts, and distribution plots, to represent complex shipment data in an intuitive manner. Research indicates that visualization-driven systems significantly enhance decision-making by enabling stakeholders to quickly identify bottlenecks and performance trends.

Furthermore, the integration of modern data processing tools such as Python, along with libraries like Pandas and Matplotlib, has improved the efficiency of handling large-scale logistics data. Web-based frameworks, including dashboarding tools, have enabled the development of user-friendly interfaces with dynamic filtering capabilities. However, many existing solutions either focus on limited metrics or lack comprehensive integration of multiple analytical features within a single platform.

Therefore, there remains a need for a unified and interactive logistics dashboard that combines data preprocessing, visualization, and real-time filtering to provide comprehensive insights. The proposed system aims to address these gaps by offering an integrated solution for efficient supply chain analysis and decision-making. In recent years, the application of data analytics and visualization techniques in supply chain management has gained significant attention. Traditional logistics systems primarily depend on static reporting methods, which often lack real-time interactivity and limit the ability

to identify operational inefficiencies. Studies have highlighted that such approaches make it difficult to monitor key performance indicators (KPIs) such as delivery time, route efficiency, and regional performance.

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III. PROBLEM STATEMENT

The rapid growth of logistics and supply chain operations has led to an exponential increase in the volume and complexity of shipment data. Organizations generate large amounts of data related to orders, shipping routes, delivery times, regions, and transportation modes. However, despite the availability of such data, many organizations struggle to effectively utilize it for performance monitoring and decision-making. Existing logistics systems often rely on static reports or fragmented tools, which provide limited insights and fail to capture the dynamic nature of supply chain operations.

One of the primary challenges faced by organizations is the lack of visibility into logistics performance across multiple dimensions. There is insufficient clarity regarding which factory-to-customer routes consistently perform efficiently and which ones are prone to delays. Without this information, it becomes difficult to identify underperforming routes and take corrective actions. Additionally, organizations lack a clear understanding of how shipping performance varies across different regions, states, and shipping modes. This absence of granular insights prevents effective comparison and optimization of logistics strategies.

Another critical issue is the inability to identify and analyze regional bottlenecks. Certain geographic areas may experience recurring delays due to factors such as infrastructure limitations, high demand, or inefficient routing. However, traditional reporting methods do not effectively highlight these spatial patterns, making it challenging for decision-makers to pinpoint

problem areas. As a result, logistics optimization efforts often remain reactive rather than proactive.

Furthermore, the lack of interactive and user-friendly analytical tools limits the accessibility of logistics insights. Many existing systems require technical expertise to interpret complex datasets, which restricts their usability for non-technical stakeholders such as managers and operations teams. The absence of dynamic filtering and real-time analysis capabilities further reduces the effectiveness of these systems in supporting timely decision-making.

Another significant problem is the difficulty in analyzing temporal trends in logistics performance. Shipping lead times may vary over time due to seasonal demand, operational changes, or external disruptions. Without proper visualization and analysis tools, it is challenging to identify such trends and anticipate potential delays. This limitation hinders the ability of organizations to plan and optimize their logistics operations effectively.

In addition, organizations often lack a unified platform that integrates multiple aspects of logistics analysis, including route efficiency, regional performance, and ship mode comparison. The absence of such an integrated system leads to fragmented insights and reduces the overall effectiveness of decision-making processes. Consequently, organizations are unable to fully leverage their data to improve operational efficiency and reduce costs.

To address these challenges, there is a need for a comprehensive, data-driven system that can provide clear, interactive, and multi-dimensional insights into logistics performance. Such a system should enable users to analyze key performance indicators, identify bottlenecks, compare shipping methods, and explore trends over time. By transforming raw data into actionable insights, the system can support proactive decision-making and contribute to the optimization of supply chain operations.

A. Identified Challenges

The development of an effective logistics performance dashboard involves several key challenges:

Data Inconsistency and Quality Issues: Logistics datasets often contain missing values, duplicates, and inconsistent formats, which affect the accuracy of analysis and require extensive preprocessing.

Lack of Real-Time Data Integration: The system primarily relies on historical data, limiting the ability to monitor live operations and respond to immediate disruptions.

Multi-Dimensional Data Complexity: Analyzing logistics performance across multiple dimensions such as region, state, route, and ship mode increases computational and visualization complexity.

Identification of Bottlenecks: Detecting geographical and operational bottlenecks from large datasets is challenging without effective visualization and aggregation techniques.

Scalability Issues: Handling large-scale logistics data efficiently while maintaining dashboard performance can be difficult as data volume increases.

User Interface and Usability: Designing an intuitive and interactive dashboard that caters to both technical and non-technical users requires careful UI/UX considerations.

Limited Predictive Capability: The current system focuses on descriptive analytics and lacks advanced predictive models for forecasting delays or optimizing routes.

B. Dataset Fields Description

TABLE I
DATASET FIELDS DESCRIPTION

Field	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date of order
Ship Date	Date of shipment
Ship Mode	Shipping method of order
Customer ID	Unique customer identifier
Country/Region	Country or region of customer
City	City of customer
State/Province	State/province of customer
Postal Code	Postal code of customer
Division	Product division
Region	Region of customer
Product ID	Unique product identifier
Product Name	Product long name
Sales	Total sales value of order
Units	Total units of order
Gross Profit	Gross profit of order (Sales - Cost)
Cost	Cost to manufacture

C. Key Performance Indicators (KPIs)

TABLE II
KEY PERFORMANCE INDICATORS (KPIs)

KPI	Description
Shipping Lead Time	Ship Date - Order Date
Average Lead Time	Mean shipping duration per route
Route Volume	Number of orders per route
Delay Frequency	% of shipments exceeding threshold
Route Efficiency Score	Normalized lead-time performance

IV. PROPOSED SYSTEM

The proposed system is an interactive Logistics Performance Dashboard designed to provide comprehensive analysis and visualization of shipping and supply chain data. The system aims to transform raw logistics data into meaningful insights by integrating data preprocessing, analytical computation, and dynamic visualization within a unified platform. It is developed to address the limitations of traditional reporting systems by enabling real-time interaction, multi-dimensional analysis, and user-friendly exploration of logistics performance.

The system follows a structured workflow beginning with data acquisition and preprocessing. Raw logistics data, including order details, shipping dates, regions, states, and ship modes, is cleaned and transformed to ensure consistency and accuracy. This includes handling missing values, removing duplicates, and converting data types such as date fields into appropriate formats. Feature engineering techniques are

applied to derive key performance indicators, such as shipping lead time, which serves as a primary metric for evaluating delivery efficiency.

Once the data is processed, the system performs exploratory data analysis to identify patterns, trends, and anomalies in logistics performance. Various analytical techniques are used to examine performance across multiple dimensions, including regional analysis, state-level performance, route efficiency, and ship mode comparison. These analyses help in uncovering insights such as high-performing regions, delayed routes, and variations in delivery times.

A key component of the proposed system is its interactive visualization module. The dashboard presents data through a variety of visual representations, including histograms, heatmaps, line charts, and geographic maps. These visualizations enable users to easily interpret complex datasets and identify important trends. For example, heatmaps are used to highlight regional bottlenecks, line charts show trends in shipping lead time over time, and choropleth maps provide a geographical perspective of state-level performance.

The system also incorporates dynamic filtering capabilities, allowing users to customize their analysis based on parameters such as date range and region. This interactivity enhances the usability of the dashboard and enables users to focus on specific areas of interest. The interface is designed to be intuitive and accessible, ensuring that both technical and non-technical users can effectively utilize the system.

The implementation of the dashboard is carried out using modern data analysis and visualization tools, ensuring scalability and performance efficiency. The system architecture supports modular development, allowing for future enhancements such as integration of real-time data sources, predictive analytics, and optimization algorithms.

Overall, the proposed system provides a comprehensive and flexible solution for logistics performance analysis. By combining data analytics with interactive visualization, it enables organizations to gain deeper insights into their operations, identify inefficiencies, and make informed decisions to optimize supply chain performance.

V. METHODOLOGY

The proposed Logistics Performance Dashboard is developed using a structured approach that involves data pre-processing, feature engineering, analytical computation, and interactive visualization. The methodology ensures efficient transformation of raw shipment data into meaningful insights.

A. Data Collection and Data Pre-Processing

The dataset consists of shipment-related attributes such as order date, shipping date, region, state, ship mode, and delivery performance. Initial preprocessing is performed to ensure data consistency and accuracy.

Missing values are handled appropriately, and date-related columns are converted into standard datetime format to enable time-based analysis. Additionally, irrelevant or inconsistent records are removed to improve data quality.

```
df["Order Date"] = pd.to_datetime(df["Order Date"])
filtered_df = df[(df["Order Date"] >= date1) & (df["Order Date"] <= date2)]
```

To enable dynamic analysis, the dataset is filtered based on user-selected date ranges and regions.

```
filtered_df = filtered_df[
    (filtered_df["Order Date"] >= pd.to_datetime(
        date1)) &
    (filtered_df["Order Date"] <= pd.to_datetime(
        date2))
]

if region:
    filtered_df = filtered_df[filtered_df["Region"].
        isin(region)]
```

B. Feature Engineering

To derive meaningful insights, new features are created from the existing dataset. A key metric, Shipping Lead Time, is calculated as the difference between shipping date and order date.

```
filtered_df["Shipping Lead Time"] = (
    filtered_df["Ship Date"] - filtered_df["Order
        Date"]
).dt.days
```

Additional grouping and aggregation operations are performed to evaluate performance across regions, states, and routes. These derived features help in identifying inefficiencies and trends.

C. Analytical Computation

The system computes multiple key performance indicators (KPIs) to assess logistics efficiency. These include:

Average shipping lead time, Route-wise performance, Region-wise shipment distribution, Delays across shipping modes.

Grouping and aggregation techniques are used to summarize data for analysis.

```
route_ranking = filtered_df.groupby('Factory to
    Customer Region')['Shipping Lead Time'].mean().
    round(2).sort_values(ascending=True).reset_index
    ()

region_congestion = filtered_df.groupby('Region').
    agg(
        Avg_Lead_Time=('Shipping Lead
            Time', 'mean'),
        Shipment_Count=('Shipping Lead
            Time', 'count'),
        Lead_Time_Std=('Shipping Lead
            Time', 'std')).round(2).
        sort_values(by='
            Avg_Lead_Time', ascending=
            False).reset_index()
```

These computations form the basis for visualization and interpretation.

D. Dashboard Development

The dashboard is implemented using Streamlit, providing an interactive user interface for real-time data exploration. The sidebar components are used to allow users to select filters such as region and date range.

```
region = st.sidebar.multiselect("Pick your Region",
                                df["Region"].unique())

st.markdown("### Select Date Range")
col1, col2 = st.columns(2)
# Getting the min and max date
startDate = pd.to_datetime(df["Order Date"]).min()
endDate = pd.to_datetime(df["Order Date"]).max()

with col1:
    date1 = pd.to_datetime(st.date_input("Start Date",
                                          startdate))

with col2:
    date2 = pd.to_datetime(st.date_input("End Date",
                                          enddate))
```

The interface dynamically updates visualizations based on user inputs, enabling flexible and intuitive analysis.

E. Data Visualization

Various visualization techniques are used to effectively represent logistics performance:

Histogram for shipping lead time distribution, Bar charts for route and regional performance, Heatmaps for identifying geographic bottlenecks,

These visualizations are generated using the Plotly, Matplotlib, and Seaborn libraries.

```
import plotly.express as px
import seaborn as sns
import matplotlib.pyplot as plt

fig1 = px.bar(
    route_ranking,
    x="Shipping Lead Time",
    y="Factory to Customer Region",
    orientation="h",
    template="plotly_dark",
    title="Average Shipping Lead Time by Route",
    color="Shipping Lead Time",
    color_continuous_scale="plasma",
    hover_data={"Shipping Lead Time": True})

fig1.update_layout(
    title_x=0.25,
    xaxis_title="Average Lead Time",
    yaxis_title="Route",
    height=350)

fig1.update_traces(
    textposition="outside")

fig2 = px.bar(
    Total_Shipments,
    x="Sales",
    y="Factory to Customer Region",
    orientation="h",
    template="plotly_dark",
    title="Route Performance by Total Sales",
    color="Sales",
    color_continuous_scale="plasma",
    hover_data={"Sales": True})
```

```
fig2.update_layout(
    title_x=0.25,
    xaxis_title="Total Sales",
    yaxis_title="Route",
    height=350)

fig2.update_traces(
    textposition="outside")

return (fig1, fig2)
```

The visual outputs help in identifying delays, trends, and inefficiencies in the logistics system.

F. System Workflow

The overall workflow of the system follows a sequential process:

1. Data collection and preprocessing
2. Feature engineering and metric computation
3. Application of filters(navigation,region,state,date,threshold)
4. Visualization and analysis through dashboard
5. Interpretation of results for decision-making

VI. RESULTS AND DISCUSSION

The developed Logistics Performance Dashboard was evaluated using shipment data to analyze key performance metrics and identify operational inefficiencies. The results are presented through various visualizations, providing insights into shipping performance, regional variations, and route efficiency.

A. Route Efficiency Overview

1) *Shipping Lead Time Analysis:* The distribution of shipping lead time is shown in Fig. 1. It can be observed that most shipments are delivered within a moderate time range, while a smaller portion experiences higher delays. This indicates variability in delivery performance across different routes.

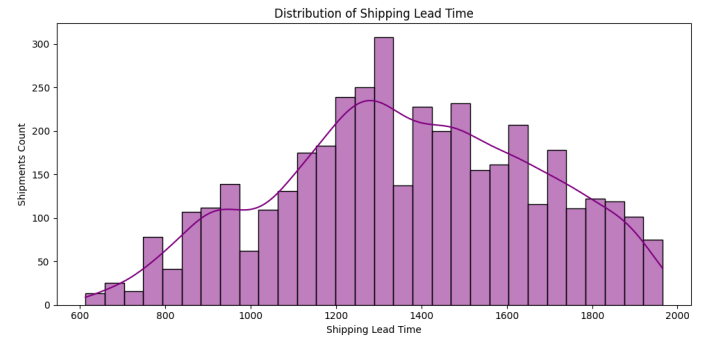


Fig. 1. Shipping Lead Time Distribution

2) *Route-wise Performance Analysis:* Route-wise performance analysis identifies the most efficient and least efficient factory-to-customer routes. Efficient routes maintain consistently lower shipping times compared to others.



Fig. 2. Route-wise Performance Analysis

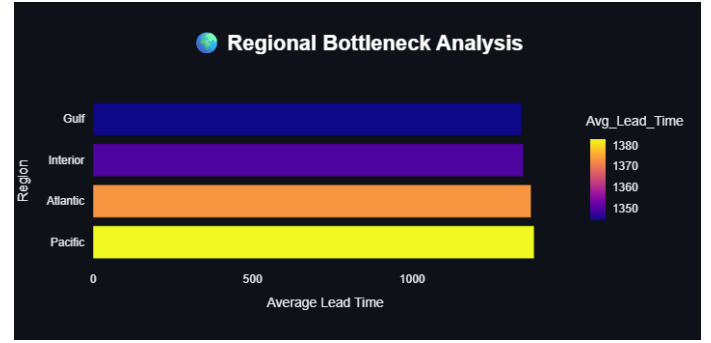


Fig. 4. Regional Bottleneck Analysis

B. Geographic Shipping Map

1) *US Shipping Performance by State:* The choropleth map visualizes shipping performance across different U.S. states using color intensity to represent key metrics such as total orders and average shipping lead time. States with higher order volumes or longer lead times are highlighted with more intense colors, enabling quick identification of high-demand areas and potential delays. This visualization helps in understanding geographic distribution patterns and detecting regions that may require operational improvements or resource optimization.

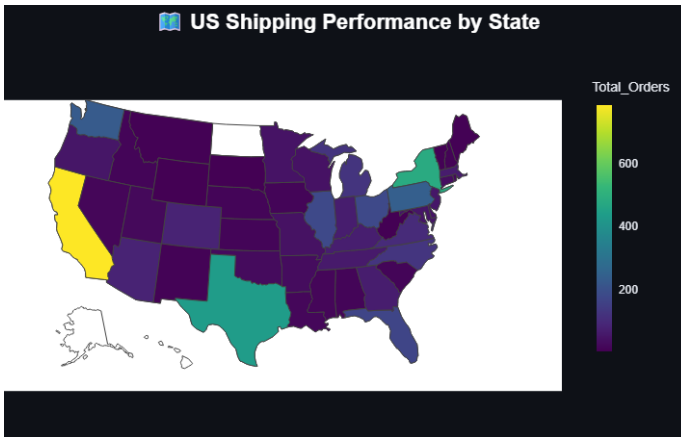


Fig. 3. US Shipping Performance by State

2) *Regional Bottleneck Analysis:* The regional bottleneck analysis identifies areas with higher concentrations of shipment delays using geographic visualization techniques. Regions exhibiting consistently higher shipping lead times are classified as bottlenecks, indicating inefficiencies in logistics operations. These delays may result from factors such as transportation constraints, infrastructure limitations, or increased demand. The analysis enables the identification of underperforming regions, allowing stakeholders to take targeted actions for improving delivery performance and optimizing resource allocation.

3) *Top 15 Performing US States (Shipping Analysis):* The heatmap presents the top 15 performing U.S. states based on shipping efficiency, using color intensity to represent average shipping lead time and order volume. States with better performance are indicated by lower lead times, while variations in color highlight differences in efficiency across regions. This visualization enables quick comparison of high-performing states and helps identify patterns in optimized logistics operations. It also assists in understanding which regions consistently achieve faster deliveries and higher operational efficiency.

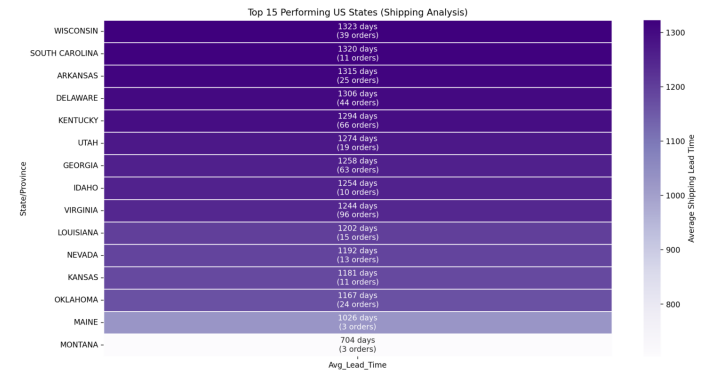


Fig. 5. Top 15 Performing US States (Shipping Analysis)

C. Ship Mode Comparison

1) *Ship Mode Comparison:* The ship mode comparison analysis evaluates the performance of different shipping methods based on average lead time. The visualization highlights variations in delivery speed across modes, indicating that certain shipping options consistently achieve faster deliveries while others exhibit longer delays. This comparison helps in identifying the most efficient shipping methods and supports decision-making for selecting optimal transport modes to improve overall logistics performance.



Fig. 6. Ship Mode Comparison



Fig. 8. Order-level shipment timelines

D. Route Drill-Down

1) *State-Level Performance Insights*: The state-level analysis of the top 15 performing regions reveals significant variation in average shipping lead time across different states. Among these, Montana demonstrates the most efficient logistics performance, exhibiting the lowest average lead time, which indicates faster and more reliable deliveries. In contrast, Arkansas records the highest average lead time within this group, suggesting potential inefficiencies or delays in its logistics operations.

The variation in performance across states may be attributed to factors such as transportation infrastructure, geographic distance, demand density, and operational constraints. States with lower lead times reflect optimized routing and efficient resource utilization, whereas higher lead times indicate possible bottlenecks or capacity limitations. This analysis enables stakeholders to benchmark state-level performance, identify areas requiring improvement, and implement targeted strategies to enhance delivery efficiency across underperforming regions.

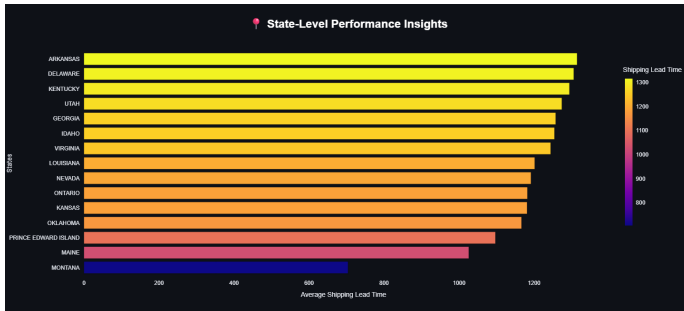


Fig. 7. State-Level Performance Insights

2) *Order-level shipment timelines*: The line chart illustrates the trend of shipping lead time over a given time period, highlighting fluctuations in delivery performance. It enables the identification of patterns such as periods of increased delays or improved efficiency. This temporal analysis helps in understanding how logistics performance evolves over time and supports proactive decision-making to address emerging issues.

VII. DISCUSSION

The analysis of logistics data through the developed dashboard provides several important insights into shipment performance, regional variations, and operational efficiency. By integrating multiple visualizations, the system enables a comprehensive understanding of patterns and inefficiencies within the supply chain.

The shipping lead time distribution reveals that while a majority of shipments are delivered within an acceptable time frame, a noticeable proportion experiences delays. This variation indicates inconsistencies in logistics operations, which may arise due to differences in route conditions, demand fluctuations, or operational constraints. The presence of outliers suggests that certain shipments are significantly delayed, highlighting the need for targeted improvements.

Regional and state-level analyses further emphasize disparities in logistics performance. Some states demonstrate consistently lower average lead times, indicating efficient transportation and resource management. In contrast, states with higher lead times reflect potential bottlenecks, possibly due to infrastructure limitations, geographic distance, or increased shipment volumes. For instance, high-performing states exhibit optimized routing and faster processing, whereas underperforming states require focused attention to enhance efficiency.

The route efficiency analysis provides deeper insight into the performance of specific factory-to-customer routes. Certain routes consistently achieve lower delivery times, making them benchmarks for optimal performance. Conversely, routes with higher delays indicate inefficiencies that may stem from poor route planning, traffic conditions, or logistical constraints. Identifying these routes allows for strategic optimization and better allocation of resources.

The comparison of different shipping modes highlights the trade-off between delivery speed and operational cost. Faster shipping modes tend to achieve lower lead times, while more economical options may result in longer delivery durations. This insight is critical for decision-making, as it enables organizations to balance cost efficiency with delivery performance based on business requirements.

The temporal analysis of shipping lead time over time reveals fluctuations in logistics performance. Periods of increased delays may correspond to peak demand, seasonal variations, or external disruptions, while stable periods indi-

cate efficient operations. Understanding these trends supports proactive planning and helps mitigate potential risks.

Additionally, the geographic visualizations, including heatmaps and choropleth maps, effectively highlight regional bottlenecks and high-demand areas. These visual tools enable quick identification of locations where delays are concentrated, facilitating targeted interventions and improved resource distribution.

Overall, the results demonstrate that the proposed dashboard successfully transforms raw logistics data into actionable insights. By enabling dynamic filtering and interactive exploration, the system enhances visibility into supply chain operations and supports data-driven decision-making. The findings emphasize the importance of integrating analytical tools in logistics management to improve efficiency, reduce delays, and optimize overall performance.

VIII. SYSTEM EVALUATION

The proposed Logistics Performance Dashboard was evaluated based on its ability to efficiently process shipment data, generate meaningful insights, and support user-driven analysis. The evaluation focuses on performance, usability, and analytical effectiveness in comparison to traditional reporting methods.

From a performance perspective, the system demonstrates efficient data handling and processing capabilities. By leveraging optimized data manipulation techniques, the dashboard is able to filter and update results dynamically based on user inputs such as region and date range. This ensures minimal latency in generating visualizations, enabling near real-time interaction with the dataset.

In terms of usability, the dashboard provides an intuitive and user-friendly interface through the use of interactive components. The inclusion of sidebar filters allows users to easily customize their analysis without requiring technical expertise. The visual representation of data through charts and maps enhances interpretability, making it easier for stakeholders to understand complex logistics patterns.

The analytical effectiveness of the system is reflected in its ability to transform raw shipment data into actionable insights. Key performance indicators such as shipping lead time, route efficiency, and regional performance are effectively captured and visualized. The system successfully identifies bottlenecks, high-performing regions, and inefficient routes, thereby supporting informed decision-making.

Compared to traditional static reporting methods, the proposed system offers significant advantages in terms of flexibility and responsiveness. While conventional approaches provide limited and delayed insights, the interactive dashboard enables real-time exploration and immediate identification of trends and anomalies.

Overall, the evaluation demonstrates that the proposed system is effective in improving visibility, enhancing analytical capabilities, and supporting data-driven optimization in logistics operations.

IX. ADVANTAGES

The proposed Logistics Performance Dashboard offers several advantages that enhance the efficiency and effectiveness of supply chain analysis. By integrating data processing, visualization, and interactivity into a single platform, the system provides significant improvements over traditional logistics monitoring approaches.

One of the primary advantages is the ability to perform real-time data analysis through dynamic filtering. Users can interactively explore shipment data based on region and date range, enabling immediate access to relevant insights without the need for manual report generation. This significantly reduces analysis time and improves responsiveness.

The system also provides enhanced data visualization, transforming complex logistics data into intuitive graphical representations such as heatmaps, histograms, and geographic maps. These visualizations simplify the interpretation of large datasets and allow users to quickly identify trends, patterns, and anomalies.

Another key advantage is the identification of operational bottlenecks. The dashboard highlights regions, states, and routes that experience higher delays, enabling targeted interventions. This supports proactive decision-making and helps organizations optimize resource allocation and logistics strategies.

The integration of multiple analytical features within a single interface ensures comprehensive performance evaluation. Metrics such as shipping lead time, route efficiency, and ship mode comparison are analyzed together, providing a holistic view of logistics operations rather than isolated insights.

Additionally, the system offers user-friendly interaction through an intuitive interface built using Streamlit. Even non-technical users can easily navigate the dashboard and perform analysis, making it accessible to a wide range of stakeholders.

Finally, the proposed system is scalable and flexible, allowing for future enhancements such as integration with real-time data sources, predictive analytics, and advanced optimization techniques. This makes it a robust solution adaptable to evolving logistics requirements.

X. LIMITATIONS

Despite its effectiveness in analyzing logistics performance, the proposed dashboard has certain limitations that must be acknowledged. Identifying these limitations is essential for understanding the scope of the system and guiding future improvements.

One of the primary limitations is the dependence on dataset quality. The accuracy and reliability of the insights generated by the dashboard are directly influenced by the completeness and correctness of the input data. Missing values, inconsistencies, or outdated records can lead to misleading interpretations and reduce analytical effectiveness.

Another limitation is the lack of real-time data integration. The current system operates on historical or preloaded datasets and does not incorporate live data from logistics operations.

As a result, it may not reflect real-time changes such as unexpected delays, traffic conditions, or operational disruptions.

The system also has limited predictive capabilities, as it primarily focuses on descriptive and exploratory analysis. While it effectively identifies existing patterns and bottlenecks, it does not provide forecasts or predictive insights, such as estimating future delays or optimizing routes using machine learning techniques.

Additionally, the dashboard is constrained by scalability and performance considerations when handling extremely large datasets. Although efficient for moderate data volumes, performance may degrade with significant increases in data size without further optimization or integration with advanced data processing frameworks.

Another limitation is the restricted scope of analysis, as the system focuses on selected metrics such as lead time, route efficiency, and regional performance. Other important factors, such as fuel costs, weather conditions, and driver availability, are not considered, which may affect the comprehensiveness of the analysis.

Finally, the system relies on a web-based interface using Streamlit, which, while user-friendly, may have limitations in customization, deployment scalability, and enterprise-level integration compared to more advanced dashboarding solutions.

XI. FUTURE SCOPE

The proposed Logistics Performance Dashboard provides a strong foundation for data-driven analysis of supply chain operations; however, there are several opportunities for further enhancement to improve its functionality, scalability, and practical applicability in real-world logistics environments.

One significant area for future development is the integration of real-time data processing capabilities. Incorporating live data streams from logistics systems, such as GPS tracking, shipment status updates, and traffic conditions, would enable the dashboard to provide up-to-date insights. This would enhance decision-making by allowing stakeholders to respond immediately to delays, disruptions, or unexpected changes in delivery schedules.

Another important extension involves the incorporation of predictive analytics using machine learning techniques. By leveraging historical shipment data, models can be developed to forecast shipping lead times, identify potential delays, and recommend optimal routes. Techniques such as regression analysis, time series forecasting, and classification models can be used to improve the accuracy of predictions and enable proactive logistics management.

The system can also be enhanced through optimization algorithms for route planning and resource allocation. Advanced techniques such as genetic algorithms or shortest path optimization can be applied to determine the most efficient routes, minimize delivery time, and reduce transportation costs. This would transform the system from a descriptive analytical tool into a prescriptive decision-support system.

Furthermore, expanding the scope of analysis to include additional influencing factors can significantly improve the

comprehensiveness of the system. Integrating data related to weather conditions, fuel costs, traffic patterns, and driver availability would provide deeper insights into the causes of delays and inefficiencies. This multi-dimensional analysis would support more accurate and informed decision-making.

Another potential improvement is the deployment of the system on cloud platforms to enhance scalability and accessibility. Cloud-based deployment would allow the dashboard to handle large-scale datasets efficiently and enable access from multiple locations, making it suitable for enterprise-level applications. Integration with cloud services can also facilitate data storage, processing, and real-time analytics.

The user interface can be further improved by incorporating advanced visualization techniques and customization features. Interactive dashboards with drill-down capabilities, customizable reports, and role-based access can enhance user experience and make the system more adaptable to different stakeholder requirements.

Finally, the system can be extended to support automated alert and notification mechanisms. By setting predefined thresholds for key performance indicators, the dashboard can automatically notify users when delays or anomalies occur. This would enable faster response times and improve operational efficiency.

In summary, the future scope of the proposed system lies in enhancing real-time capabilities, integrating predictive and optimization techniques, expanding analytical dimensions, and improving scalability and user experience. These advancements would significantly strengthen the system's ability to support intelligent, efficient, and proactive logistics management.

XII. CONCLUSION

This paper presented the design and development of an interactive Logistics Performance Dashboard aimed at improving the analysis and monitoring of supply chain operations. The system integrates data preprocessing, analytical computation, and visualization techniques to transform raw shipment data into meaningful and actionable insights. By leveraging modern data analysis tools and an intuitive user interface, the proposed solution addresses the limitations of traditional static reporting methods.

The dashboard enables comprehensive evaluation of key performance indicators such as shipping lead time, route efficiency, regional performance, and ship mode comparison. Through various visualizations, including histograms, heatmaps, line charts, and geographic maps, the system effectively highlights patterns, trends, and operational inefficiencies. The ability to dynamically filter data based on region and date enhances analytical flexibility and allows users to explore logistics performance in a more interactive and efficient manner.

The results obtained from the analysis demonstrate that significant variations exist across regions, states, and routes. Certain areas exhibit optimized performance with lower lead times, while others indicate the presence of bottlenecks and

delays. These insights are critical for identifying inefficiencies and implementing targeted improvements in logistics operations. Additionally, the comparison of different shipping modes provides valuable understanding of the trade-offs between delivery speed and operational considerations.

The system evaluation further confirms that the proposed dashboard offers improved usability, faster data processing, and enhanced decision-making capabilities compared to conventional approaches. By providing a unified platform for data visualization and analysis, the system enhances transparency and supports data-driven strategies in supply chain management.

Despite certain limitations, such as the lack of real-time data integration and predictive capabilities, the proposed system establishes a strong foundation for advanced logistics analytics. The identified future enhancements, including machine learning integration and real-time monitoring, can further expand its applicability and effectiveness.

In conclusion, the developed Logistics Performance Dashboard successfully demonstrates the importance and impact of data-driven tools in modern logistics management. By enabling efficient analysis, identifying bottlenecks, and supporting informed decision-making, the system contributes to improved operational performance and optimized supply chain processes.

XIII. REFERENCES

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